🧠 APTITUDE MASTER FORMULA & TRICKS SHEET

📚 *Covers: Quantitative + Logical + Reasoning + Shortcuts + Examples*
🎯 *For exams: TCS | Infosys | Wipro | SSC | Bank | CAT | Govt | Freelancing aptitude tests*

~BY PYTHONBOYZ

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🔢 1. NUMBERS & DIVISIBILITY

- ➕ **Even = 2n**
- ➖ **Odd = 2n + 1**
- 🔁 **Sum of n natural numbers = n(n + 1)/2**
- ⚡ **Sum of first n even numbers = n(n + 1)**
- 🔹 **Sum of first n odd numbers = n²**
- ✳️ **Product of two numbers = HCF × LCM**
- 🔢 **Divisibility Rules:**
  - 2 → Last digit even
  - 3 → Sum of digits ÷ 3
  - 4 → Last 2 digits ÷ 4
  - 5 → Last digit 0 or 5
  - 6 → ÷2 and ÷3
  - 8 → Last 3 digits ÷8
  - 9 → Sum of digits ÷9
  - 11 → (Sum odd pos − Sum even pos) ÷11

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💰 2. PERCENTAGE / PROFIT & LOSS

- 📉 **% = (Part / Whole) × 100**
- ⚡ **Successive % change = a + b + (ab / 100)**
- 💵 **Profit % = (Profit / CP) × 100**
- 📊 **Loss % = (Loss / CP) × 100**

- 💥 **Double Discount = x + y − (xy / 100)**
- 🧮 **Profit or Loss by CP/SP:**
  - SP = CP × (100 + Profit%) / 100
  - CP = SP × 100 / (100 + Profit%)

🕐 3. TIME, SPEED & DISTANCE

- 🚙 **Speed = Distance / Time**
- ⏳ **Time = Distance / Speed**
- 🏞️ **Distance = Speed × Time**
- ⚡ **Average Speed = (2xy) / (x + y)**
- 🏁 **Relative Speed:**
  - Opposite direction = A + B
  - Same direction = |A − B|
- 🚆 **If 2 trains cross each other:**
  Time = (Sum of lengths) / (Relative Speed)

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⚙️ 4. TIME & WORK

- 🧠 **Work = Rate × Time**
- 🔹 **If A alone = x days, B alone = y days → Together = (xy) / (x + y)**
- 🧱 **Efficiency ∝ 1 / Time**
- 💪 **If A twice as efficient as B → Time ratio 1:2**
- ⏱️ **Work left = 1 − (Days worked / Total time)**

⚖️ 5. RATIO, MIXTURE & ALLIGATION

- 📏 **Ratio = a : b → Fraction = a / b**
- 🔹 **Proportion: a/b = c/d → ad = bc**
- 🧃 **Alligation Rule:**
  Ratio = (Mean − Low) : (High − Mean)
- 🤝 **Partnership Profit ∝ Capital × Time**

💸 6. INTEREST (SIMPLE & COMPOUND)

- 💰 **SI = (P × R × T) / 100**
- 💎 **CI = P[(1 + R/100)^n − 1]**
- ⚡ **Difference (CI − SI) for 2 years = P × (R/100)²**
- 📉 **Amount = P + SI | Amount = P(1 + R/100)^n**

📐 7. GEOMETRY & MENSURATION

- 🔺 **Triangle = ½ × base × height**
- ⚫ **Circle = πr²**
- 🔲 **Square = a² | Rectangle = l × b**
- 🧊 **Cube: Volume = a³ | TSA = 6a² | Diagonal = a√3**
- 🧯 **Cuboid: Volume = l × b × h | Diagonal = √(l² + b² + h²)**
- 🧩 **Cylinder: Volume = πr²h | CSA = 2πrh**
- 🔺 **Cone: Volume = ⅓πr²h | Slant height = √(r² + h²)**
- 🌀 **Sphere: Volume = (4/3)πr³ | Area = 4πr²**

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🧮 8. ALGEBRA & PROGRESSION

- 📈 **$(a + b)^2 = a^2 + b^2 + 2ab$**
- 📉 **$a^2 - b^2 = (a + b)(a - b)$**
- 🔹 **$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$**
- 📊 **AP (Arithmetic Progression):**
  - nth term = $a + (n - 1)d$
  - Sum = $n/2\ [2a + (n - 1)d]$
- 📉 **GP (Geometric Progression):**
  - nth term = ar^{n-1}
  - Sum = $a(r^n - 1)/(r - 1)$

🎲 9. PROBABILITY & PERMUTATION

- 🎯 **Probability = Favourable / Total**
- 🧩 **nPr = n! / (n − r)!**
- 🔢 **nCr = n! / [r!(n − r)!]**
- ⚡ **nCr = nC(n−r)**
- 🎰 **n! = n × (n−1) × (n−2) × ... × 1**

📊 10. LOGARITHMS & INDICES

- 🔹 **$\log_a(MN) = \log_a M + \log_a N$**
- ➖ **$\log_a(M/N) = \log_a M - \log_a N$**

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- ⚡ **$\log_a(M^k) = k \times \log_a M$**
- 🔁 **$\log_a b = 1 / \log_ba$**
- 💥 **$(a^m)(a^n) = a^{(m + n)}$**
- 🧮 **$(a^m)/(a^n) = a^{(m - n)}$**
- 💡 **$(a^m)^n = a^{(mn)}$**

🧠 11. LOGICAL REASONING TRICKS

- ⏰ **Clock Angle = |30H − (11/2)M|**
- 🗓️ **Leap Year = Divisible by 4 but not 100, or divisible by 400**
- 🧭 **Direction:**
  - Left turn = 90° anti-clockwise
  - Right turn = 90° clockwise
- 👨‍👩‍👧 **Blood Relation:**
  - Father's brother = Uncle
  - Sister's son = Nephew
  - Brother's daughter = Niece
- 🧩 **Seating Trick:**
  - Facing center → left = clockwise
  - Facing out → left = anticlockwise

💥 12. SHORTCUT TRICKS

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- 🔢 **Square ending with 5 ⇒ (n × (n + 1)) + "25"**
    - Example: 35² = (3×4)25 = 1225
- ⚡ **×5 ⇒ Half ×10 | ÷5 ⇒ ×2 ÷10**
- 💯 **Percent ⇄ Fraction:**
    - 50% = ½
    - 25% = ¼
    - 20% = ⅕
    - 33⅓% = ⅓
    - 12½% = ⅛
- 🧠 **If A is twice efficient as B → Time ratio 1 : 2**
- 🔁 **Net % effect = a + b + (ab/100)**

📄 13. EXAMPLES (SHORT & CLEAN)

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- 🧱 **Work:**
  A = 10 days, B = 15 days
  → 1 day = 1/10 + 1/15 = 1/6 ⇒ Total = 6 days
- 🚙 **Average Speed:**
  60 & 40 ⇒ (2×60×40)/(60+40) = 48 km/h
- 💸 **Profit-Loss Combo:**
  +20% & −20% ⇒ Net = (20 − 20 − (20×20/100)) = −4%
- 💥 **Square of 95:**
  (9×10)25 = 9025

💀 PART 2:
ADVANCED SHORTCUTS + TRAPS

🎯 *Covers: DI, Logical, Pattern, Puzzle, Quant, Company-style Tricks (TCS | Infosys | Wipro | Deloitte | SSC | CAT)*

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📊 1. DATA INTERPRETATION (DI) SHORTCUTS

- 🔹 **% Change = (New − Old) / Old × 100**
- ⚡ **Average of 5 items = Total ÷ 5**
- 🧠 **Ratio conversion in DI:**
    - If 3:2 = 5 parts,
      → 60 students ⇒ each part = 60/5 = 12
      → A = 36, B = 24
- 💥 **Quick Average Trick:**
    - If 3 values differ equally (ex: 40, 50, 60) ⇒ Middle value = average
- 📉 **Approximation hack:**
    - (99 × 47) ≈ (100×47) − 47 = 4653

🧮 2. SIMPLIFICATION / APPROXIMATION

- ⚡ **BODMAS Rule** → Bracket → Of → Division → Multiplication → Addition → Subtraction
- 💥 **(x + a)(x + b) = x² + (a + b)x + ab**
- 🔹 **Vedic Trick:**
    - For 99 × 99 = (100−1)(100−1) = 10000 − 200 + 1 = 9801
- 🧠 **Square root close estimate:**
    - √50 ≈ 7.07 | √60 ≈ 7.74 | √80 ≈ 8.94
- 📄 **Cube close estimate:**
    - $21^3 = (20+1)^3 = 8000 + 1200 + 60 + 1 = 9261$

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💸 3. PROFIT / DISCOUNT / PARTNERSHIP ADVANCED

- 💰 **Successive Discount Trick:**
  - Two discounts x%, y% → Net = x + y − (xy/100)
- 🔥 **Equal Gain/Loss → SP = Mean Proportional ($\sqrt{(CP_1 \times CP_2)}$)**
- 🤝 **Partnership Ratio:**
  Profit ∝ (Investment × Time)
  - $A : B = (x_1 \times t_1) : (x_2 \times t_2)$
- 💎 **Compound Interest Double Formula:**
  $T = (\log(2) / \log(1 + R/100))$

🕐 4. TIME, WORK & PIPE TRICKS

- 💦 **Pipe Inlet & Outlet Trick:**
  - Net work = (Inlet +ve) − (Outlet −ve)
- ⚙️ **If A 50% more efficient than B → Time ratio 2 : 3**
- 🧱 **If 3 workers: A(10), B(15), C(20) → Total = 1/10+1/15+1/20 = 1/4.62 ⇒ 4.62 days**
- ⏱️ **Work & Wages Relation:**
  Wages ∝ Work done

🚙 5. SPEED, TRAIN, BOAT, RELATIVE MOTION

- 🚆 **Train Crossing:**
  - Speed = (Distance / Time) × 18/5 (m/s to km/h)
- 🌊 **Boat in Stream:**
  - Downstream = u + v | Upstream = u − v

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- Speed in still water = (Down + Up) / 2
- Stream speed = (Down − Up) / 2

- 🧭 **Relative Motion:**
  - Opposite = sum | Same = diff

- ⚡ **Chasing problem:**
  Distance / (Speed diff) = Time to catch

🧠 6. REASONING — PATTERN + PUZZLES

🔹 Series Pattern

- 2, 6, 12, 20, 30 → +4, +6, +8, +10 → next +12 = 42
- 1, 4, 9, 16 → Squares → next 25
- 3, 9, 27 → ×3 → next 81

🔹 Coding-Decoding

- CAT → 3120 (C=3, A=1, T=20)
- **Reverse trick:** ABC → 123 → reversed → CBA = 321

🔹 Direction

- N → E = +90° (Right turn)
- N → W = −90° (Left turn)
- One full round = 360°

🔹 Seating

- Facing center → Left = Clockwise

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- Facing out → Left = Anti-clockwise

📈 7. PERMUTATION / COMBINATION SHORT TRICKS

- 🔢 **nPr = n! / (n − r)!**
- 🔹 **nCr = n! / [r!(n − r)!]**
- 💥 **nCr = nC(n−r)**
- ⚡ **If all letters distinct:**
  - WORD of 4 letters = 4! = 24
- 🔁 **If repetition exists:**
  - Formula = n! / (p! q! …)

🧮 8. AGE & AVERAGE SHORTCUTS

- 👶 **Present age ratio → multiply with total years to get diff**
  Example: A:B = 3:5, diff 10 years → 2 parts = 10 → 1 part = 5 → A=15, B=25
- 📊 **Average formula:**
  Avg = (Sum of obs) / (No. of obs)
- 🔹 **When 1 new item added:**
  New Avg = Old Avg + (Diff / n+1)

🧩 9. LOGICAL PUZZLE SHORTCUTS

- 🔁 **If two statements contradict → Either–Or rule**
- 💡 **If A always true ⇒ B follows**
  (→ means direct inference)

- ⚖️ **Syllogism Trick:**
  - All + All = All
  - All + Some = Some
  - Some + No = No conclusion

🔥 10. SEATING / ARRANGEMENT TIPS

- 🔹 *Circle arrangement:*
  - Facing center → Left = clockwise
  - Facing out → Left = anti-clockwise
- 🧠 *Linear arrangement:*
  - "To the left of" → left side of your view
  - "Immediate right of" → next seat right

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💥 11. DATA SUFFICIENCY / DECISION MAKING

- ✅ **Statement 1 sufficient, 2 not → (A)**
- ✅ **Statement 2 sufficient, 1 not → (B)**
- ✅ **Both needed → (C)**
- ❌ **Neither sufficient → (D)**
- 💡 *Always check independent data first*

⚙️ 12. QUANT SHORTCUT MIX

- 🧿 **(x% of y) = (y% of x)**
- 🔹 **$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$**
- ⚡ **If sum constant → product max when all equal**
- 🧠 **Simple trick:**
  - 1% of X = X / 100
  - 0.5% of X = X / 200

📚 13. COMPANY APTITUDE PATTERN INSIGHT

Company	Question Focus	Key Topics	Time
TCS NQT	Reasoning + Quant	Time-Speed, DI, Pattern	75 min
Infosys	Logical Heavy	Arrangement, Data	60 min
Wipro	Arithmetic Mix	Ratio, CI, Probability	60 min
Deloitte	Logical Puzzles	Syllogism, Series	90 min
Accenture	DI + English	Graphs, Ratios	90 min

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💎 14. SUPER TRICKS (TOP 10 QUICK SOLVERS)

1. 🔢 **Square of 95 ⇒ (9×10)25 = 9025**
2. ⚡ **$(a+b)^2-(a-b)^2 = 4ab$**
3. 🚗 **Avg speed = 2xy/(x+y)**
4. 💰 **Profit +20%, −20% ⇒ Net −4%**
5. 📈 **(n−1)th term = a + (n−2)d**
6. 💥 **$(x^3 + y^3) = (x+y)(x^2 - xy + y^2)$**
7. 💡 **LCM × HCF = Product of numbers**
8. 🧮 **$\sqrt{(a^2 + b^2)}$ for diagonal or hypotenuse**
9. 💸 **CI for 2 yrs = SI + $P(R/100)^2$**
10. 🧠 **Percent → Fraction table (remember forever)**

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💀 PART 3:
EXAM STRATEGY + TIME TRAPS

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🧭 1️⃣ OVERALL STRATEGY (UNIVERSAL RULES)

- 🧠 **Rule 1 – Speed > Accuracy > Guesswork**
- 🕒 **Rule 2 – 30 sec rule:** if you can't see the path in 30 sec, SKIP it.
- 💀 **Rule 3 – No emotional attachment**: don't waste time "just because you started."
- 🔁 **Rule 4 – Pattern first → Formula second → Solve third.**
- 🧮 **Rule 5 – Solve easy→medium→hard order.**

🎯 2️⃣ COMPANY WISE APTITUDE PATTERN & TRICKS

💼 TCS NQT

- 🕐 **Time:** 75 min | 26 Qs
- 🔹 Focus = Arithmetic + Logic + Speed Math
- 💥 *Hot Topics:* Time & Work, Ages, DI, Series, Speed
- ⚙️ **Tip:** Use ratio instead of fractions.
- ⚡ **Trap:** "More than twice" ≠ × 2 — it means × 3.

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🧠 INFOSYS

- 🕒 **Time:** 60 min | 10 Qs
- 💡 *Focus:* Puzzles & Reasoning
- 🧩 *Types:* Circular arrangement, Data links, Family relation
- ⚠️ **Trap:** Facing inside/outside reversal kills accuracy.
- ⚡ **Trick:** Draw tables → eliminate false options quickly.

💻 WIPRO

- 🕑 **Time:** 60 min | 15 Qs
- 🎯 *Focus:* Percentages, CI, Speed, Probability
- ⚙️ **Shortcut:** Convert everything to ratio form.
- ⚠️ **Trap:** Decimals camouflage ratios — simplify ×100.

🏢 DELOITTE

- 🕑 **Time:** 90 min | 20 Qs
- 💡 *Focus:* Logical Puzzle + Inference
- 🔹 *Type:* Statement vs Assumption, Syllogism, Decision
- ⚠️ **Trap:** “Cannot be true” = Reverse logic.
- ⚡ **Trick:** Use Venn diagram for Syllogism.

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⚙️ ACCENTURE

- 🕓 **Time:** 90 min | 30 Qs
- 📈 *Focus:* DI + Ratio + Tables + English mix
- ⚡ **Shortcut:** Estimate by eye then approx.
- 💥 **Trap:** Graphs with similar scales → check axis labels.

🧮 SSC CGL / BANK

- 📚 *Pattern:* Quant Heavy (Algebra, Trig, Mensuration)

- ⚙️ **Trick:** Use “Smart Substitution” for long Algebra.
- ⚡ **Trap:** They change variables to confuse you (x→y or a→b).

🎓 **CAT / MBA**

- 🧠 *Focus:* Logical DI, High-level Quant, Verbal Reasoning
- ⚙️ **Trick:** Option Elimination > Full Solution
- ⚡ **Trap:** “Data Insufficient” options often true (30% cases).

🧩 3️⃣ TOPIC-WISE TIME STRATEGY

Topic	Avg Solve Time	Best Trick
Numbers	40 sec	Divisibility rules + mod logic
Ratio & Proportion	50 sec	Cross-multiplication
Time & Work	60 sec	LCM method for days
Speed & Distance	45 sec	2xy/(x+y) formula
Percent & Profit	40 sec	Successive formula
DI	60–80 sec	Approximation

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Logical Puzzles	2–3 min	Table elimination
Coding-Decoding	30 sec	Letter position rule

💣 4️⃣ TIME-SAVING TACTICS

- ⏱️ **Vedic Math Hacks:**
  - *By 9's:* To find remainder mod 9 → Sum of digits
  - *By 11's:* Alternate sum difference method
- 💥 **Square Shortcut:**
  - If ends with 5 → n(n+1) + 25
- 💡 **Cross Multiplication for Ratio:**
  - a:b = 3:5 and b:c = 4:7 → a:b:c = 12:20:35
- 🧠 **Elimination Rule:**
  - Plug in simple numbers like 100 for % problems.
- ⚡ **Graph DI:**
  - Read Title → Units → Base → Type before calculating.

🧠 5️⃣ PSYCHOLOGICAL TRAPS IN EXAMS

- ⚠️ *They test your time control, not your math depth.*
- 💭 **First 2 Qs = Confidence setters** → pick easy ones.
- 🧩 **Options are reverse-engineered** → check for same digits pattern.

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- 🧠 **“All of these”** = correct only ~20% of time.
- 💀 **“None of these”** = correct ~30% in bank papers.

🔥 6️⃣ FINAL FORMULA RECALL (QUICK REVISION BLOCK)

- **$(a+b)^2 = a^2+b^2+2ab$**
- **Speed = D/T, Work = Rate×Time**
- **SI = (PRT)/100**
- **CI – SI = $P(R/100)^2$ (for 2 yrs)**
- **$\log_a b = 1/\log_ba$**
- **LCM × HCF = Product**
- **P(E)=Favourable/Total**
- **Sum of AP = n/2 [2a+(n−1)d]**
- **Angle (Clock) = |30H – (11/2)M|**

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🏁 7️⃣ ENDGAME CHECKLIST

✅ Mastered Formulas
✅ Understood Tricks
✅ Practiced 10 DI sets
✅ Timed yourself (< 60 sec/Q)
✅ Analyzed wrong answers
✅ Re-took mock under pressure